

Music Genre Classification

Using Logistic Regression,
KNN and PCA+KNN

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Problem and Dataset

Goal: Predict the top-level genre of a music track from precomputed audio features.

- Dataset: Free Music Archive (FMA) metadata and feature tables.
- Files Used: tracks.csv, genres.csv, and features.csv.
- Final working subset: 6,746 tracks, 8 genres, 518 features.
- Genres Used: Hip-Hop, Pop, Rock, Experimental, Folk, Jazz, Electronic, Soul-RnB.



Methods

Logistic Regression

Linear multi-class
baseline

KNN

Predicts genre using
nearest neighbors in
feature space

PCA + KNN

Proposed hybrid
method using PCA
before KNN to reduce
dimensionality.

Motivation for PCA + KNN: KNN can struggle in high-dimensional spaces.

Experimental Setup

80/20 stratified train-test split.

StandardScaler applied in pipelines for fair comparison.

```
Shape after genre filtering: (43595, 518)
Number of tracks per genre:
(track, genre_top)
Rock          14182
Experimental   10608
Electronic     9372
Hip-Hop        3552
Folk           2803
Pop            2332
Jazz           571
Soul-RnB       175
Name: count, dtype: int64
Final X_small shape: (6746, 518)
Final y_small distribution:
(track, genre_top)
Hip-Hop        1000
Pop            1000
Rock           1000
Experimental    1000
Folk           1000
Electronic     1000
Jazz           571
Soul-RnB       175
Name: count, dtype: int64
```

Experimental Setup

KNN test ed with $k = 3, 5, 7, 9$.

```
print("\nBest KNN k:", best_k)
print("Best KNN accuracy:", best_knn_acc)
```

```
KNN accuracy (k=3): 0.4822222222222222
KNN accuracy (k=5): 0.49925925925925924
KNN accuracy (k=7): 0.5044444444444445
KNN accuracy (k=9): 0.49333333333333335
```

```
Best KNN k: 7
Best KNN accuracy: 0.5044444444444445
```

Experimental Setup

PCA + KNN tested with 20, 50,
and 100 components; $k = 3, 5, 7$.

```
print('Best PCA+KNN accuracy: ', best_pca_knn_acc)
```

```
PCA+KNN accuracy (n_components=20, k=3): 0.4318518518518519  
PCA+KNN accuracy (n_components=20, k=5): 0.43925925925925924  
PCA+KNN accuracy (n_components=20, k=7): 0.45925925925925926  
PCA+KNN accuracy (n_components=50, k=3): 0.4792592592592593  
PCA+KNN accuracy (n_components=50, k=5): 0.4962962962962963  
PCA+KNN accuracy (n_components=50, k=7): 0.4911111111111111  
PCA+KNN accuracy (n_components=100, k=3): 0.48444444444444446  
PCA+KNN accuracy (n_components=100, k=5): 0.5022222222222222  
PCA+KNN accuracy (n_components=100, k=7): 0.5044444444444445
```

```
Best PCA+KNN settings: (100, 7)
```

```
Best PCA+KNN accuracy: 0.5044444444444445
```

Results

	Best Setting	Accuracy
»» Logistic Regression	Default multinomial setup	0.5319
»» KNN	$k = 7$	0.5044
»» PCA + KNN	100 components, $k = 7$	0.5044



Logistic regression performed best overall.

Logistic Regression accuracy: 0.5318518518518518



PCA + KNN matched KNN but did not improve on it.

Best KNN accuracy: 0.5044444444444445

Best PCA+KNN accuracy: 0.5044444444444445

Conclusion

	Method	Test Accuracy
0	Logistic Regression	0.545926
1	KNN (k=7)	0.494815
2	PCA + KNN (100 components, k=7)	0.500741

- Linear logistic regression worked better than local neighbor-based methods on this feature set.
- PCA + KNN was a reasonable proposed improvement, but it did not outperform KNN.

Future Work

- Try SVM or random forest.
- Use per-class precision/recall and confusion matrices.
- Explore spectrogram-based or deep learning approaches.



Thanks

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